



Machine Learning - Gradient Boosting



Gradient Boosting Machines (GBM) is a powerful machine learning technique that is widely used for building predictive models. It is a type of ensemble method that combines the predictions of multiple weaker models to create a stronger and more accurate model.

GBM is a popular choice for a wide range of applications, including regression, classification, and ranking problems. Let's understand the workings of GBM and how it can be used in machine learning.

What is a Gradient Boosting Machine (GBM)?

GBM is an iterative machine learning algorithm that combines the predictions of multiple decision trees to make a final prediction.

The algorithm works by training a sequence of decision trees, each of which is designed to correct the errors of the previous tree.

In each iteration, the algorithm identifies the samples in the dataset that are most difficult to predict and focuses on improving the model's performance on these samples.

This is achieved by fitting a new decision tree that is optimized to reduce the errors on the difficult samples. The process continues until a specified stopping criteria is met, such as reaching a certain level of accuracy or the maximum number of iterations.

How Does a Gradient Boosting Machine Work?

The basic steps involved in training a GBM model are as follows –

- **Initialize the model** – The algorithm starts by creating a simple model, such as a single decision tree, to serve as the initial model.

- **Calculate residuals** – The initial model is used to make predictions on the training data, and the residuals are calculated as the differences between the predicted values and the actual values.
- **Train a new model** – A new decision tree is trained on the residuals, with the goal of minimizing the errors on the difficult samples.
- **Update the model** – The predictions of the new model are added to the predictions of the previous model, and the residuals are recalculated based on the updated predictions.
- **Repeat** – Steps 3-4 are repeated until a specified stopping criteria is met.

GBM can be further improved by introducing regularization techniques, such as L1 and L2 regularization, to prevent overfitting. Additionally, GBM can be extended to handle categorical variables, missing data, and multi-class classification problems.

Example

Here is an example of implementing GBM using the Sklearn breast cancer dataset –

```
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.ensemble import GradientBoostingClassifier
from sklearn.metrics import accuracy_score

# Load the breast cancer dataset
data = load_breast_cancer()
X = data.data
y = data.target

# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)

# Train the model using GradientBoostingClassifier
model = GradientBoostingClassifier(n_estimators=100, max_depth=3,
learning_rate=0.1)
model.fit(X_train, y_train)

# Make predictions on the testing set
y_pred = model.predict(X_test)

# Evaluate the model's accuracy
```

```
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

Output

In this example, we load the breast cancer dataset using Sklearn's `load_breast_cancer` function and split it into training and testing sets. We then define the parameters for the GBM model using `GradientBoostingClassifier`, including the number of estimators (i.e., the number of decision trees), the maximum depth of each decision tree, and the learning rate.

We train the GBM model using the `fit` method and make predictions on the testing set using the `predict` method. Finally, we evaluate the model's accuracy using the `accuracy_score` function from Sklearn's `metrics` module.

When you execute this code, it will produce the following output –

```
Accuracy: 0.956140350877193
```

Advantages of Using Gradient Boosting Machines

There are several advantages of using GBM in machine learning –

- **High accuracy** – GBM is known for its high accuracy, as it combines the predictions of multiple weaker models to create a stronger and more accurate model.
- **Robustness** – GBM is robust to outliers and noisy data, as it focuses on improving the model's performance on the most difficult samples.
- **Flexibility** – GBM can be used for a wide range of applications, including regression, classification, and ranking problems.
- **Interpretability** – GBM provides insights into the importance of different features in making predictions, which can be useful for understanding the underlying factors driving the predictions.
- **Scalability** – GBM can handle large datasets and can be parallelized to accelerate the training process.

Limitations of Gradient Boosting Machines

There are also some limitations to using GBM in machine learning –

- **Training time** – GBM can be computationally expensive and may require a significant amount of training time, especially when working with large datasets.
- **Hyperparameter tuning** – GBM requires careful tuning of hyperparameters, such as the learning rate, number of trees, and maximum depth, to achieve optimal performance.
- **Black box model** – GBM can be difficult to interpret, as the final model is a combination of multiple decision trees and may not provide clear insights into the underlying factors driving the predictions.

TOP TUTORIALS

[Python Tutorial](#)

[Java Tutorial](#)

[C++ Tutorial](#)

[C Programming Tutorial](#)

[C# Tutorial](#)

[PHP Tutorial](#)

[R Tutorial](#)

[HTML Tutorial](#)

[CSS Tutorial](#)

[JavaScript Tutorial](#)

[SQL Tutorial](#)

TRENDING TECHNOLOGIES

[Cloud Computing Tutorial](#)

[Amazon Web Services Tutorial](#)

[Microsoft Azure Tutorial](#)

[Git Tutorial](#)

[Ethical Hacking Tutorial](#)

[Docker Tutorial](#)

[Kubernetes Tutorial](#)

[DSA Tutorial](#)

[Spring Boot Tutorial](#)

[SDLC Tutorial](#)

[Unix Tutorial](#)

CERTIFICATIONS

[Business Analytics Certification](#)
[Java & Spring Boot Advanced Certification](#)
[Data Science Advanced Certification](#)
[Cloud Computing And DevOps](#)
[Advanced Certification In Business Analytics](#)
[Artificial Intelligence And Machine Learning](#)
[DevOps Certification](#)
[Game Development Certification](#)
[Front-End Developer Certification](#)
[AWS Certification Training](#)
[Python Programming Certification](#)

COMPILERS & EDITORS

[Online Java Compiler](#)
[Online Python Compiler](#)
[Online Go Compiler](#)
[Online C Compiler](#)
[Online C++ Compiler](#)
[Online C# Compiler](#)
[Online PHP Compiler](#)
[Online MATLAB Compiler](#)
[Online Bash Terminal](#)
[Online SQL Compiler](#)
[Online Html Editor](#)

[ABOUT US](#) | [OUR TEAM](#) | [CAREERS](#) | [JOBS](#) | [CONTACT US](#) | [TERMS OF USE](#) |
[PRIVACY POLICY](#) | [REFUND POLICY](#) | [COOKIES POLICY](#) | [FAQ'S](#)





Tutorials Point is a leading Ed Tech company striving to provide the best learning material on technical and non-technical subjects.

© Copyright 2025. All Rights Reserved.